Package 'surfscape'

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Type Package

Title Continuous Landscape Structure and Its Change Through Time

Version 0.2.0

Description Tools for separating compositional from configurational change in continuous landscape surfaces such as vegetation indices, canopy cover or productivity. Provides an exact three-way decomposition of the change in any moving-window metric into a shift in the landscape value distribution, a spatial redistribution of values among neighbourhoods and a rearrangement of values within neighbourhoods, together with an audit of whether a metric is able to respond to spatial arrangement at all. Built on 'terra' with a 'Rcpp' window engine covering surface, texture, autocorrelation, entropy and lacunarity metrics.

License GPL-3

Encoding UTF-8

LazyData true

Depends R (>= 4.1)

Imports terra (>= 1.7), Rcpp (>= 1.0.10), stats

LinkingTo Rcpp

Suggests testthat (>= 3.0.0), sf, exactextractr, geodiv, GLCMTextures, rasterdiv, spdep, knitr, rmarkdown, ggplot2

RoxygenNote 7.3.1

Config/testthat/edition 3

URL https://github.com/ehsanrahimi666/surfscape

BugReports https://github.com/ehsanrahimi666/surfscape/issues

NeedsCompilation yes

VignetteBuilder knitr

R topics documented:

ss_decompose 2
ss_extract 4
ss_metrics 5
ss_metric_list 6
ss_rank_transfer 7
ss_scale_profile 8
ss_sensitivity 9
ss_sensitivity_fun 10

Index **13**

`ss_decompose` *Decompose change in a continuous landscape surface*

Description

Splits the change in any continuous landscape metric between two dates into three additive components: a shift in the landscape-wide value distribution, a spatial redistribution of values among neighbourhoods, and a rearrangement of values within neighbourhoods. The three components sum exactly to the observed change at every cell.

Usage

```
ss_decompose(
  z1,
  z2,
  metrics = c("sq", "sr1"),
  w = 7,
  components = c("total", "global", "local", "config"),
  range = NULL,
  nbins = 32L,
  nlevels = 16L,
  lacbox = 3L,
  filename = "",
  overwrite = FALSE,
  chunk_rows = NULL,
  wopt = list()
)
```

Arguments

`z1, z2`	Single-layer ‘SpatRaster‘ objects for the two dates, with identical geometry. Cells missing in either date are dropped from both.
`metrics`	Character vector of metric names, see [ss_metric_list()].
`w`	Window size in cells: an odd integer or ‘c(nrow, ncol)‘.

`components` Which components to return. Any subset of ‘"total"‘, ‘"global"‘, ‘"local"‘, ‘"config"‘.

`range` Value range shared by both dates, used for binning. Defaults to the range of the two dates combined, which is the correct choice for a temporal comparison.

`nbins, nlevels, lacbox`
Metric settings, see [ss_metrics()].

`filename, overwrite, chunk_rows, wopt`
Output and blocking control.

Value

A ‘SpatRaster‘ with one layer per metric and component, named ‘<metric>_<component>‘.

The decomposition

Let M be a metric evaluated on a neighbourhood, W_1 and W_2 the neighbourhood at the two dates, and let A denote the rank transfer that keeps one arrangement and substitutes the other value multiset. Symmetrising over the two possible orderings (the two-factor Shapley solution) gives

$$\Delta M = \underbrace{\Delta_{\text{global}}}_{\text{distribution}} + \underbrace{\Delta_{\text{local}}}_{\text{redistribution}} + \underbrace{\Delta_{\text{config}}}_{\text{arrangement}}$$

where the arrangement component is obtained by transferring ranks *within* each window, the global component by transferring ranks over the *whole* landscape, and the local component as the remainder. The identity holds exactly, not approximately: if the two dates differ only by a monotone transformation of values the arrangement and local components are exactly zero, and if they share an identical value multiset the global component is exactly zero.

For every metric with ‘arrangement = FALSE‘ in [ss_metric_list()] the configurational component is exactly zero by construction, at every scale. That is the point: such metrics cannot detect spatial rearrangement, however large it is.

See Also

[ss_sensitivity()] to audit a metric before using it, [ss_scale_profile()] to run the decomposition across scales.

Examples

```
set.seed(1)
z1 <- terra::rast(nrows = 40, ncols = 40)
terra::values(z1) <- stats::rnorm(1600)
z1 <- terra::focal(z1, 5, "mean")
z2 <- terra::rast(z1)
terra::values(z2) <- stats::rnorm(1600)
z2 <- terra::focal(z2, 9, "mean")
d <- ss_decompose(z1, z2, metrics = c("sq", "sr1"), w = 7)
names(d)
```

`ss_extract` *Extract decomposed change around point locations at several radii*

Description

Summarises the output of [ss_decompose()] in circular buffers around field sites, at several radii, and returns a tidy long table ready for multi-scale modelling.

Usage

```
ss_extract(x, pts, radii, fun = mean, coords = c("x", "y"), id = NULL)
```

Arguments

`x`	A ‘SpatRaster‘, typically the output of [ss_decompose()].
`pts`	An ‘sf‘ object, a ‘SpatVector‘ of points, a two-column matrix, or a ‘data.frame‘ with coordinate columns.
`radii`	Numeric vector of buffer radii in map units.
`fun`	Summary function applied within each buffer.
`coords`	Column names holding x and y when ‘pts‘ is a ‘data.frame‘.
`id`	Optional column name in ‘pts‘ to carry through as a site identifier.

Value

A long ‘data.frame‘ with columns ‘site‘, ‘radius‘, ‘layer‘, ‘value‘.

Buffer geometry

Buffers are circular and defined by radius, so buffer area scales with ‘radius^2‘. When comparing extraction shapes rather than sizes, match on area rather than on a linear dimension, otherwise any apparent effect of shape is confounded with a threefold difference in the area sampled.

Examples

```
set.seed(1)
z1 <- terra::rast(nrows = 50, ncols = 50, xmin = 0, xmax = 500,
                  ymin = 0, ymax = 500)
terra::values(z1) <- stats::rnorm(2500)
z2 <- terra::rast(z1); terra::values(z2) <- stats::rnorm(2500)
d <- ss_decompose(z1, z2, metrics = "sr1", w = 5)
pts <- data.frame(x = c(100, 300), y = c(100, 400))
ss_extract(d, pts, radii = c(50, 100))
```

ss_metrics *Moving-window metrics for continuous landscape surfaces*

Description

Computes continuous landscape metrics in a moving window. Every metric is derived from a single gather of each neighbourhood, and the neighbourhood is held as a two-dimensional patch so that configuration-sensitive metrics (within-window autocorrelation, slope, lacunarity, grey-level co-occurrence textures) are available alongside the aspatial amplitude and information-theoretic metrics.

Usage

```
ss_metrics(
  x,
  metrics = c("sq", "ssk", "sr1", "gcon"),
  w = 9,
  range = NULL,
  nbins = 32L,
  nlevels = 16L,
  lacbox = 3L,
  filename = "",
  overwrite = FALSE,
  chunk_rows = NULL,
  wopt = list()
)
```

Arguments

x	A single-layer ‘SpatRaster‘, or anything [terra::rast()] accepts.
metrics	Character vector of metric names; see [ss_metric_list()].
w	Window size: a single odd integer, or ‘c(nrow, ncol)‘, in cells.
range	Numeric length-2 value range used to bin ‘sent‘, ‘srao‘ and the grey levels of the GLCM metrics. Defaults to the range of ‘x‘. **Supply the same ‘range‘ for both dates when comparing metrics through time**, otherwise the binning differs between dates and the comparison is not meaningful; [ss_decompose()] does this automatically.
nbins	Number of bins for ‘sent‘ and ‘srao‘.
nlevels	Number of grey levels for the GLCM metrics.
lacbox	Gliding-box side length for ‘slac‘.
filename, overwrite, chunk_rows, wopt	Output and blocking options.

Value

A ‘SpatRaster‘ with one layer per requested metric, in the order given by ‘metrics‘.

Relationship to other packages

geodiv implements a wider set of surface metrics, including the bearing-area, spectral and correlation-length families, and **GLCMTextures** implements grey-level co-occurrence textures with more options; both should be preferred when those specific metrics are the goal, and both can be passed to [ss_sensitivity()] as user functions. The kernel here exists because [ss_decompose()] must evaluate the same metric on counterfactual neighbourhoods that never exist as a raster and therefore cannot be handed to an external package.

Examples

```
f <- system.file("extdata", "ndvi_2013.tif", package = "surfscape")
if (nzchar(f)) {
  r <- terra::rast(f)
  m <- ss_metrics(r, c("sq", "sr1", "gcon"), w = 9)
  names(m)
}
```

---

ss_metric_list *Catalogue of continuous landscape metrics*

Description

Lists the metrics computed by the **surfscape** window kernel together with the property the decomposition framework depends on: whether a metric responds to the *arrangement* of values inside the analysis neighbourhood, or only to the *multiset* of values it contains.

Usage

```
ss_metric_list()
```

Details

A metric belongs to the **composition** family if permuting the values within a neighbourhood leaves it unchanged. Composition metrics can only ever measure what values are present, never how they are organised; no spatial rearrangement, however extreme, will move them. This is a property of the definition rather than of the data, and it can be verified for any metric, including metrics from other packages, with [ss_sensitivity()].

Metrics belonging to the **configuration** family respond to arrangement and are therefore the only ones capable of detecting a change in spatial structure at a fixed value distribution.

The amplitude family of surface metrics (Sa, Sq, Ssk, Sku, S10z) is aspatial, as noted in passing by McGarigal et al. (2009), Frazier and Wang (2016) and Kedron et al. (2018); so are the information-theoretic measures (Shannon entropy, Rao's quadratic entropy), because they are computed from the value distribution alone. The consequences of this for change detection are what [ss_decompose()] addresses.

Value

A ‘data.frame‘ with columns ‘metric‘, ‘id‘, ‘family‘, ‘arrangement‘ and ‘definition‘.

References

McGarigal K, Tagil S, Cushman SA (2009) Surface metrics: an alternative to patch metrics for the quantification of landscape structure. *Landscape Ecology* 24:433-450.

Haralick RM, Shanmugam K, Dinstein I (1973) Textural features for image classification. *IEEE Transactions on Systems, Man, and Cybernetics* 3:610-621.

Allain C, Cloitre M (1991) Characterizing the lacunarity of random and deterministic fractal sets. *Physical Review A* 44:3552-3558.

Examples

```
ss_metric_list()
subset(ss_metric_list(), arrangement)     # configuration-sensitive only
```

---

ss_rank_transfer *Global rank transfer*

Description

Builds a counterfactual surface that keeps the spatial rank ordering of ‘arrangement‘ and takes the value distribution of ‘values‘. Only cells that are non-missing in both inputs are used.

Usage

```
ss_rank_transfer(arrangement, values)
```

Arguments

`arrangement, values`
: Single-layer ‘SpatRaster‘ objects with identical geometry.

Value

A ‘SpatRaster‘.

Examples

```
r1 <- terra::rast(nrows = 20, ncols = 20)
r2 <- terra::rast(nrows = 20, ncols = 20)
set.seed(1)
terra::values(r1) <- stats::rnorm(400)
terra::values(r2) <- stats::rnorm(400, 2)
ss_rank_transfer(r2, r1)
```

`ss_scale_profile` *Scale profile of decomposed landscape change*

Description

Runs [ss_decompose()] over a ladder of window sizes and summarises how the three components of change scale. The scale at which the configurational component peaks answers a question the total change cannot: at what grain did the landscape actually rearrange.

Usage

```
ss_scale_profile(
  z1,
  z2,
  metrics = c("sq", "sr1"),
  w_seq = c(3, 7, 15, 31),
  range = NULL,
  nbins = 32L,
  nlevels = 16L,
  lacbox = 3L,
  rasters = FALSE,
  progress = TRUE
)
```

Arguments

`z1, z2`	Single-layer 'SpatRaster' objects for the two dates, with identical geometry. Cells missing in either date are dropped from both.
`metrics`	Character vector of metric names, see [ss_metric_list()].
`w_seq`	Vector of odd window sizes in cells.
`range`	Value range shared by both dates, used for binning. Defaults to the range of the two dates combined, which is the correct choice for a temporal comparison.
`nbins, nlevels, lacbox`	Metric settings, see [ss_metrics()].
`rasters`	Logical. If 'TRUE' the decomposition rasters for each scale are returned alongside the summary.
`progress`	Logical, print progress.

Value

A 'data.frame' with one row per window size and metric, containing the mean absolute magnitude of each component, the configurational share, the correlation between components across cells, and the proportion of cells where the sign of the total change differs from the sign of the configurational change. When 'rasters = TRUE', a list with elements 'summary' and 'rasters'.

Examples

```
set.seed(1)
z1 <- terra::rast(nrows = 60, ncols = 60)
terra::values(z1) <- stats::rnorm(3600)
z1 <- terra::focal(z1, 5, "mean")
z2 <- terra::rast(z1)
terra::values(z2) <- stats::rnorm(3600)
z2 <- terra::focal(z2, 9, "mean")
ss_scale_profile(z1, z2, metrics = "sr1", w_seq = c(5, 9))
```

---

ss_sensitivity *Audit how much a metric responds to spatial arrangement*

Description

Quantifies the share of a metric's variation that comes from the spatial arrangement of values rather than from the values themselves. Windows are sampled from a real surface, the metric is evaluated on the observed arrangement and on random permutations of the same values, and the two are compared.

Usage

```
ss_sensitivity(
  x,
  metrics = ss_metric_list()$metric,
  w = 9,
  n = 1000,
  nperm = 99,
  seed = NULL,
  range = NULL,
  nbins = 32L,
  nlevels = 16L,
  lacbox = 3L
)
```

Arguments

x	A single-layer 'SpatRaster'.
metrics	Character vector of metric names; defaults to every metric in [ss_metric_list()].
w	Window size in cells.
n	Number of windows to sample.
nperm	Number of within-window permutations per sampled window.
seed	Optional integer seed.
range, nbins, nlevels, lacbox	Metric settings, see [ss_metrics()].

Details

'csi' separates two questions that the literature has treated as one. The 'arrangement' column states whether a metric *can* respond to configuration; 'csi' states how much it *does* in a particular landscape. A metric with 'arrangement = FALSE' always returns 'csi = 0' and 'effect = 0' exactly, which is a useful check that the audit is behaving.

Value

A 'data.frame' with one row per metric:

csi configurational sensitivity index, $1 - R^2$ of the observed metric regressed on its permutation mean. Zero means the metric is entirely determined by the value multiset and cannot see arrangement; one means arrangement explains all of its variation.

effect mean absolute difference between the observed value and the permutation mean, in units of the between-window standard deviation.

exceed proportion of windows whose observed value falls outside the central 95 percent of its own permutation distribution.

arrangement whether the metric can respond to arrangement at all, as a matter of construction.

Examples

```
set.seed(1)
r <- terra::rast(nrows = 60, ncols = 60)
terra::values(r) <- stats::rnorm(3600)
r <- terra::focal(r, 5, "mean")
ss_sensitivity(r, metrics = c("sq", "sr1", "gcon"), w = 9,
               n = 200, nperm = 19, seed = 1)
```

---

ss_sensitivity_fun *Audit any metric function for configurational sensitivity*

Description

The open counterpart of [ss_sensitivity()]. Instead of the metrics built into **surfscape**, this audits an arbitrary R function that maps a neighbourhood to one or more numbers, which makes it possible to test metrics implemented in other packages (**geodiv**, **GLCMTextures**, **rasterdiv**) or defined by the user on exactly the same footing.

Usage

```
ss_sensitivity_fun(
  x,
  fun,
  w = 9,
  n = 300,
  nperm = 99,
  seed = NULL,
```

```
  complete_only = TRUE,
  max_tries = 50L
)
```

Arguments

x	A single-layer 'SpatRaster'.
fun	A function taking a 'w x w' numeric matrix and returning a numeric vector. If the vector is named, the names are used as metric names; otherwise names are generated. The function must return a vector of the same length for every window.
w	Window size in cells (single odd integer, or 'c(nrow, ncol)').
n	Number of windows to sample.
nperm	Number of within-window permutations per sampled window.
seed	Optional integer seed.
complete_only	Logical. Sample only windows with no missing values (default 'TRUE'), which is what most external metric implementations require.
max_tries	Maximum number of candidate windows examined while searching for 'n' complete ones.

Details

The permutation reshuffles the values inside the neighbourhood and leaves the multiset of values untouched. Any metric that is a function of that multiset alone is therefore exactly invariant, and both 'csi' and 'effect' are zero to machine precision. The audit makes no assumption about how the metric is implemented, so a published implementation can be tested directly rather than reasoned about from its formula.

Value

A 'data.frame' with the same columns as [ss_sensitivity()]: 'metric', 'n', 'csi', 'effect', 'exceed', plus 'invariant', which is 'TRUE' when the observed value equalled the permuted value in every window to within numerical tolerance. 'invariant = TRUE' is the operational definition of an aspatial metric: it cannot detect spatial rearrangement, in this landscape or any other.

See Also

[ss_sensitivity()] for the built-in metrics, which is much faster because it avoids the R callback.

Examples

```
set.seed(1)
r <- terra::rast(nrows = 80, ncols = 80)
terra::values(r) <- stats::rnorm(6400)
r <- terra::focal(r, 5, "mean")

f <- function(m) {
  v <- as.numeric(m)
```

```
  d <- v - mean(v)
  c(sd = sqrt(mean(d^2)),                              # aspatial
    lag1 = mean((m[, -ncol(m)] - m[, -1])^2))          # spatial
}
ss_sensitivity_fun(r, f, w = 9, n = 100, nperm = 19, seed = 1)
```

Index

ss_decompose, 2
ss_extract, 4
ss_metric_list, 6
ss_metrics, 5
ss_rank_transfer, 7
ss_scale_profile, 8
ss_sensitivity, 9
ss_sensitivity_fun, 10